

**PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)**

II B.Tech II Semester Regular Examinations, April - 2025

DATABASE MANAGEMENT SYSTEMS

(Common to CSE, IT, CSE(AI&ML), CSE(AI), CSE(DS), CSE(CS))

Time: 3 hours

Max. Marks: 70

Note:

- i. Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.
- ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

Q. No.	Questions	BTL	CO	Marks
1.	a) List out some representative application of databases.	*K1	CO1	2M
	b) What is a schema?	K1	CO1	2M
	c) What is a foreign key? Give an Example.	K2	CO2	2M
	d) What is the use of embedded SQL?	K2	CO2	2M
	e) What is the purpose of the WHERE clause in a SELECT statement?	K2	CO3	2M
	f) Define a view. How does an updatable view differ from a non-updatable view?	K2	CO3	2M
	g) What is the purpose of normalization in database design?	K2	CO4	2M
	h) Differentiate between 1NF and 2NF.	K2	CO4	2M
	i) Define a transaction in a database system.	K2	CO5	2M
	j) Mention two differences between lock-based and timestamp-based concurrency protocols.	K2	CO5	2M
UNIT – I				
2.	a) Illustrate the advantages of using a database system over a traditional file system in managing records.	K3	CO1	5M
	b) Demonstrate the client-server database architecture.	K3	CO1	5M
OR				
3.	a) Develop an Entity Relationship model for a library management system. Clearly State the problem Definition, Description, Business Rule and any assumption you make.	K3	CO1	5M
	b) Write a note on representation of entities in Entity Relationship Model.	K2	CO1	5M
UNIT – II				
4.	a) Explain the importance of the null values with examples.	K2	CO2	5M
	b) Discuss about Relational Algebra.	K2	CO2	5M
OR				
5.	a) Write a command that is used for a table creation. Explain how constraints are specified in SQL during table creation with example.	K3	CO2	5M
	b) Discuss the role of DML operations in maintaining data in a relational database.	K2	CO2	5M

UNIT – III					
6.	a)	Demonstrate how to implement key constraints and integrity constraints using the CREATE TABLE	K3	CO3	5M
	b)	Explain Date and Time functions with examples.	K2	CO3	5M
OR					
7.	a)	Explain the process of ordering query results using ORDER BY. Show examples of ascending and descending order sorting.	K3	CO3	5M
	b)	Write SQL queries to implement different types of joins.	K2	CO3	5M
UNIT – IV					
8.	a)	Define functional dependency and explain its role in identifying normal forms.	K2	CO4	5M
	b)	What is dependency preserving decomposition? Why is it important?	K2	CO4	5M
OR					
9.	a)	Describe the importance of multivalued dependencies (MVD) in relational databases.	K2	CO4	5M
	b)	Explain loss less join decomposition with an example.	K2	CO4	5M
UNIT – V					
10.	a)	Explain the ACID properties of a transaction with suitable examples.	K2	CO5	5M
	b)	Explain the concept of recoverability in transactions. What are cascading rollbacks?	K2	CO5	5M
OR					
11.	a)	Explain the need for recovery algorithms. How do they ensure atomicity?	K2	CO5	5M
	b)	Discuss about hash-based indexing.	K2	CO5	5M